

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

A batch of banana milkshakes consists of **4** cups of ice cream and **2** bananas and has **1,114 milligrams (mg)** of calcium. There is **276 mg** of calcium in **1** cup of the ice cream used to make this batch of milkshakes. How much calcium, **in mg**, is in **1** banana?

Answer

- A. **5**
- B. **10**
- C. **419**
- D. **1,104**

Correct Answer: A

Rationale

Choice A is correct. It’s given that a batch of banana milkshakes consists of **4** cups of ice cream and **2** bananas and has **1,114** mg of calcium. It’s also given that there is **276** mg of calcium in **1** cup of the ice cream used to make this batch of milkshakes. It follows that the total amount of calcium in **4** cups of ice cream is **4(276)**, or **1,104** mg. Let ***x*** represent the amount of calcium, in mg, in **1** banana. It follows that the total amount of calcium in **2** bananas is **2*x*** mg. Since the batch of banana milkshakes has **1,114** mg of calcium, the equation **1,104 + 2*x* = 1,114** represents this situation. Subtracting **1,104** from both sides of this equation yields **2*x* = 10**. Dividing both sides of this equation by **2** yields ***x* = 5**. Therefore, the amount of calcium in **1** banana is **5** mg.

Choice B is incorrect. This is the amount of calcium, in mg, in **2** bananas, not in **1** banana.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the amount of calcium, in mg, in **4** cups of ice cream, not in **1** banana.